Incentive Design with Spillovers

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Michigan

Holmstrom’s appliance



Holmstrom’s appliance

Model from Holmstrom “Moral hazard & observability” (79)



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Holmstrom (79) solved this problem, giving the first-order conditions for optimal allocation of payments across outcomes.



Holmstrom’s new appliance



Holmstrom's new appliance

Setting: Team repairs an appliance through costly efforts $a_i \geq 0$.

Their performance $Y(a_1, a_2, \ldots, a_n)$ — repair quality — isn't contractible.

What is contractible: Observable **outcome** s, stochastically increasing in $Y(a_1, a_2, \ldots, a_n)$.



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We don't know how to allocate outcome-contingent incentives optimally across agents.

This is despite Holmstrom (82) on "Moral hazard in teams" and much related work.



Literature

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Moral hazard, e.g.:

Holmstrom (79), *Moral hazard and observability*

Holmstrom (82), *Moral hazard in teams*, Mookherjee (84), Legros and Matsushima (91), Legros and Matthews (93), and many others

Bernstein and Winter (12), *Contracting with Heterogeneous Externalities*, Halac, Winter, and Kremer (20)

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Our contribution: multi-agent model with stochastic outcomes as in Holmstrom (79).

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How optimal incentive allocation across agents depends on productivity **spillovers** between agents.

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Claveria Mayol, Milán, and Oviedo-Dávila (24): *Incentive Contracts and Peer Effects in the Workplace*, Sun and Zhao (24): *Contracts on Peer Pressure Networks*

Our contribution: endogenize incentives flexibly.

We leverage ideas from this literature without the strong parametric assumptions typically made there.

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refine beta

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KRISHNA DASARATHA, BENJAMIN GOLUB, AND ANANT SHAH


#### Abstract

A principal uses payments conditioned on stochastic outcomes of a team project to elicit costly effort from the team members. We develop a multi-agent generalization of a classic first-order approach to contract optimization by leveraging methods from network games. The main results characterize the optimal allocation of incentive pay across agents and outcomes. Incentive optimality requires equalizing, across agents, a product of (i) individual productivity (ii) organizational centrality and (iii) responsiveness to monetary incentives.


[^0]
## 1. Introduction

A popular method of motivating members of a team is giving them performance incentives that depend on jointly achieved outcomes. Examples of such incentives include startup executives receiving firm stock and a marketing team receiving bonuses for achieving a sales target. How should such incentive schemes be designed and how should they take into account the team's production function?

We examine these questions in a simple non-parametric model of a team working on a joint project. Each member chooses how much costly effort to exert. These actions jointly determine a real-valued team performance—for example, the quality of a product—according to a nice, increasing function of the efforts, which may entail interactions such as complementarities among agents' efforts. Any performance level
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Positivity of centrality in Lemma 10

All that remains to show is that $\widetilde{c}_{i}>0$ for every agent. This follows from (40): $\widetilde{\boldsymbol{S}}$ is non-negative and every element of $\widetilde{\boldsymbol{\alpha}}$ is strictly positive.

The proof of Lemma 10 argues that the centrality vector $\widetilde{\boldsymbol{c}}$ is positive because it is the product of a positive vector $\widetilde{\boldsymbol{\alpha}}$ and the matrix sum $\sum_{\ell=0}^{\infty} \widetilde{\boldsymbol{S}}^{\ell}$, which is claimed to be non-negative on the grounds that "$\widetilde{\boldsymbol{S}}$ is non-negative". However, the definition of $\widetilde{\boldsymbol{S}}$ involves the Hessian matrix $\widetilde{\boldsymbol{G}}$, which can have negative entries if some agents' efforts are substitutes (a possibility allowed by the model). If $\widetilde{\boldsymbol{S}}$ has negative entries, the matrix inverse $(\boldsymbol{I}-\widetilde{\boldsymbol{S}})^{-1}$ is not guaranteed to be non-negative, and the conclusion that $\widetilde{\boldsymbol{c}}>0$ does not follow from this argument. Since the positivity of centrality is used in the proof of Proposition 10, this point is crucial and seems to require either a restriction on the production technology or an alternative proof.

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Team **performance** $Y(\boldsymbol{a})$ is **one-dimensional**, a function of individual costly efforts $a_i \geq 0$…

e.g., quality of a joint product

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finitely many

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Changing agent's contract affects his effort, which affects others' incentives.

Model: Binary outcome

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Agents have utilities

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$$\mathcal{U}_i(a_i, \tau_i) = U_i(\tau_i) - C_i(a_i),$$

where τ_i is money;

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$s = 1$ is associated with…

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Principal designs a **contract**

outcome-contingent payments $\tau_i \geq 0$ in case of success

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Fact

It is without loss of optimality for the principal to pay only in case of success.

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$s \in S$ ~~v_s~~

a revenue ~~v_s~~ for principal

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$Y(\boldsymbol{a})$ increasing in each a_i, twice diff'ble.

Neither $\boldsymbol{a}$ nor $Y(\boldsymbol{a})$ is contractible.

$|S|$

Two contractible **outcomes**. Outcome

$s = 1$ is associated with…

$s \in S$ v_s

a revenue for principal

probability $P(Y)$,

Principal designs a **contract**

outcome-contingent payments

$\tau_i \geq 0$ in case of success

Model: Binary outcome

Agents have utilities

$$\mathcal{U}_i(a_i, \tau_i) = U_i(\tau_i) - C_i(a_i),$$

concave, increasing — U_i; convex, increasing — C_i

where τ_i is money;

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$\tau_{i,s} \geq 0$

Principal’s problem

Principal’s problem

Principal **knows** the environment

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Principal **knows** the environment

parameters

U_i — utility function of money

C_i — cost function of effort

Y — team performance

$P_s(\cdot)$ — probability of outcome s as a function of team performance

v_s — revenue at outcome s

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If there is no pure-strategy equilibrium given contract $\boldsymbol{\tau}$, principal gets 0.

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We focus on the binary special case. General results are similar.

Principal's problem

Principal **knows** the environment

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$U_i(\tau_i) = \tau_i$

C_i — cost function of effort

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$U_i(\tau_i) = \tau_i$

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This problem has an **intensive** margin (how much to pay) and an **extensive** margin (whom to pay). We start w/ **intensive**.

Informal statement of main result

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Our main result will show that contract optimality is characterized by a certain **index being equalized** across all agents receiving incentive pay.

Defining productivity and centrality

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Consider any equilibrium $\boldsymbol{a}^*$ given $\boldsymbol{\tau}$.

Define i's ***productivity*** as

$$\alpha_i = \frac{\partial Y}{\partial a_i}(\boldsymbol{a}^*)$$

Note marginal product can depend on others' efforts.

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Technical assumption

Work with a $\boldsymbol{\tau}$ such that $\boldsymbol{a}^*(\boldsymbol{\tau})$ is differentiable in $\boldsymbol{\tau}$.

We extend Rogerson (85) on sufficient conditions for the first-order approach.

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To define **centralities**, consider a perturbed game:

$$\mathcal{U}_i(a_i, \tau_i) = U_i(\tau_i) - C_i(a_i) + \delta_i a_i$$

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Centrality is derivative of equilibrium team performance in i's marginal returns to effort.

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$$i\text{'s } \textbf{\textit{centrality}} \text{ is } \; c_i = \frac{\mathrm{d}Y(\boldsymbol{a}^*)}{\mathrm{d}\delta_i}$$

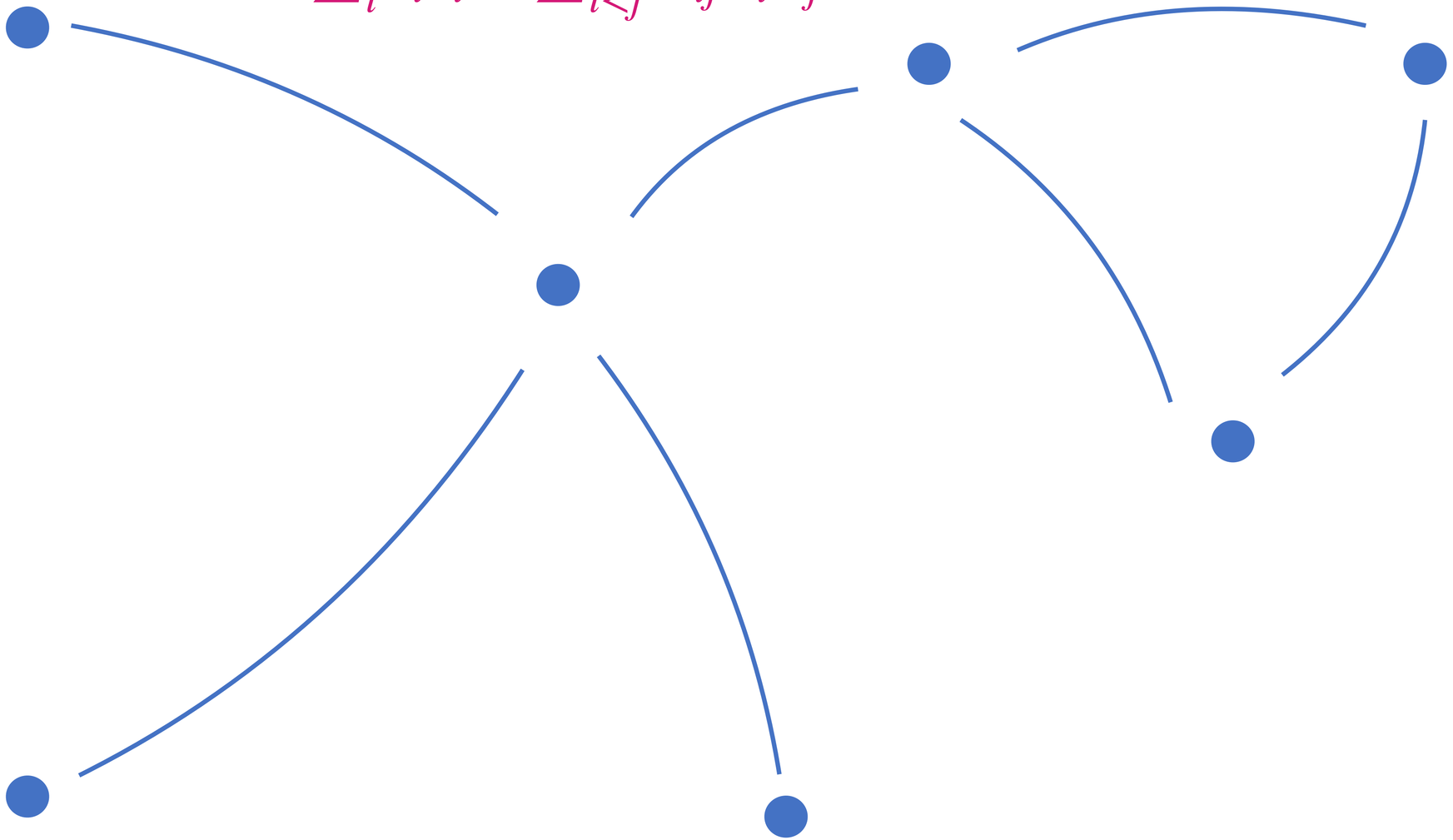
Centrality is derivative of equilibrium team performance in i's marginal returns to effort.

Reflects aggregate effects (inclusive of all spillovers) of giving i stronger incentives.

How an impulse to a player's δ_i propagates

e.g.

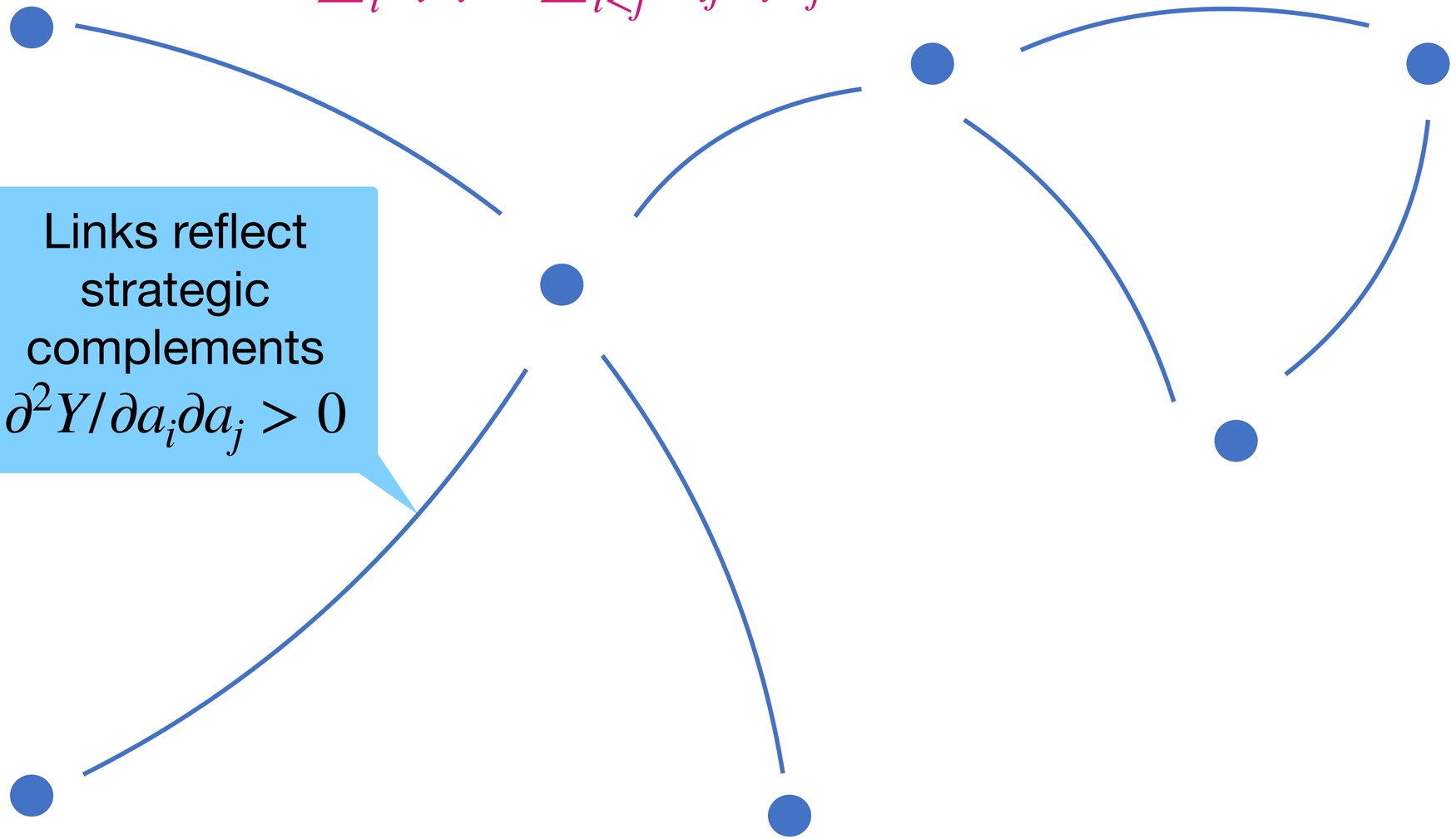
$$Y(\boldsymbol{a}) = \sum_i k_i a_i + \sum_{i<j} G_{ij}\, a_i\, a_j$$



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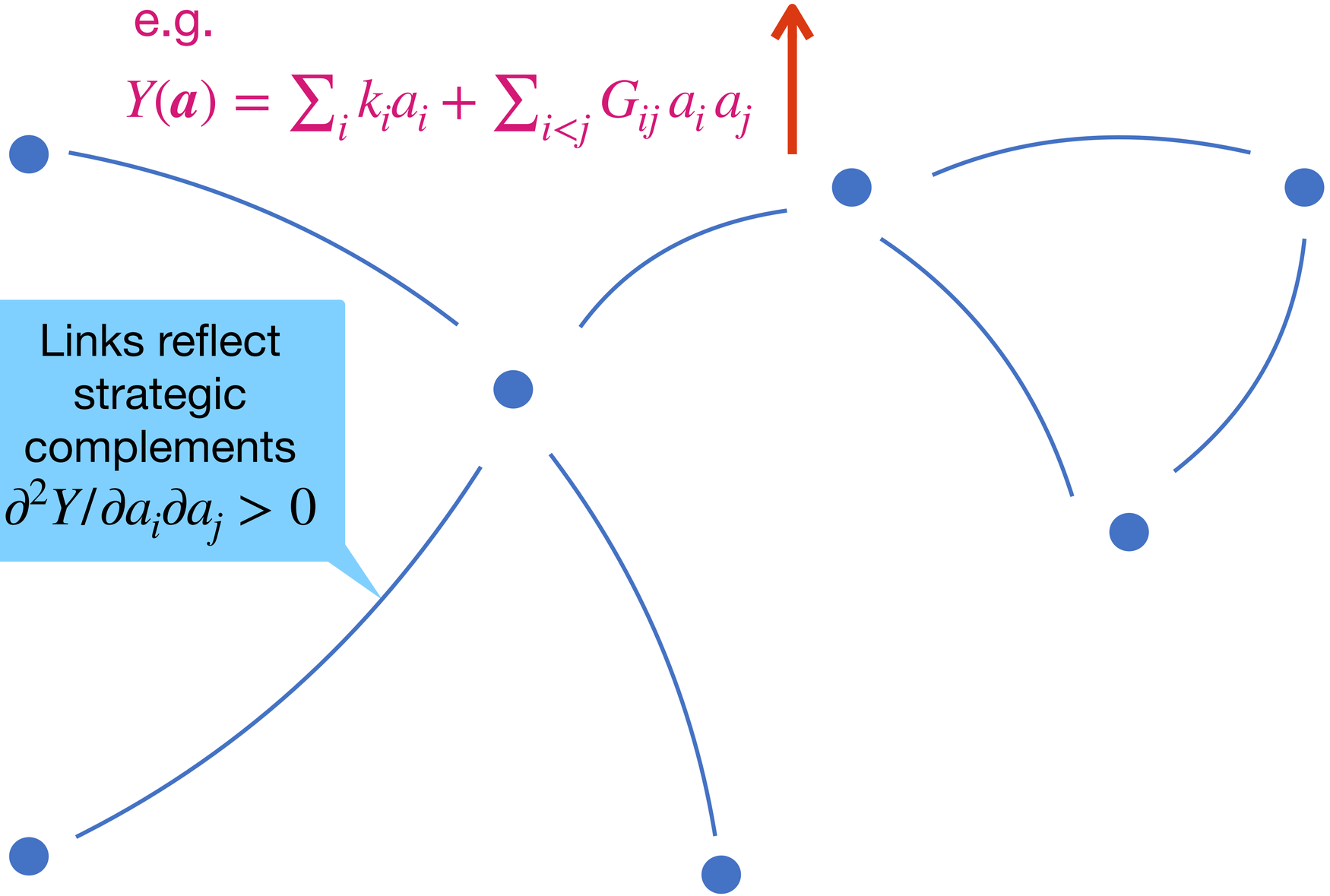
Links reflect strategic complements

$\partial^2 Y / \partial a_i \partial a_j > 0$

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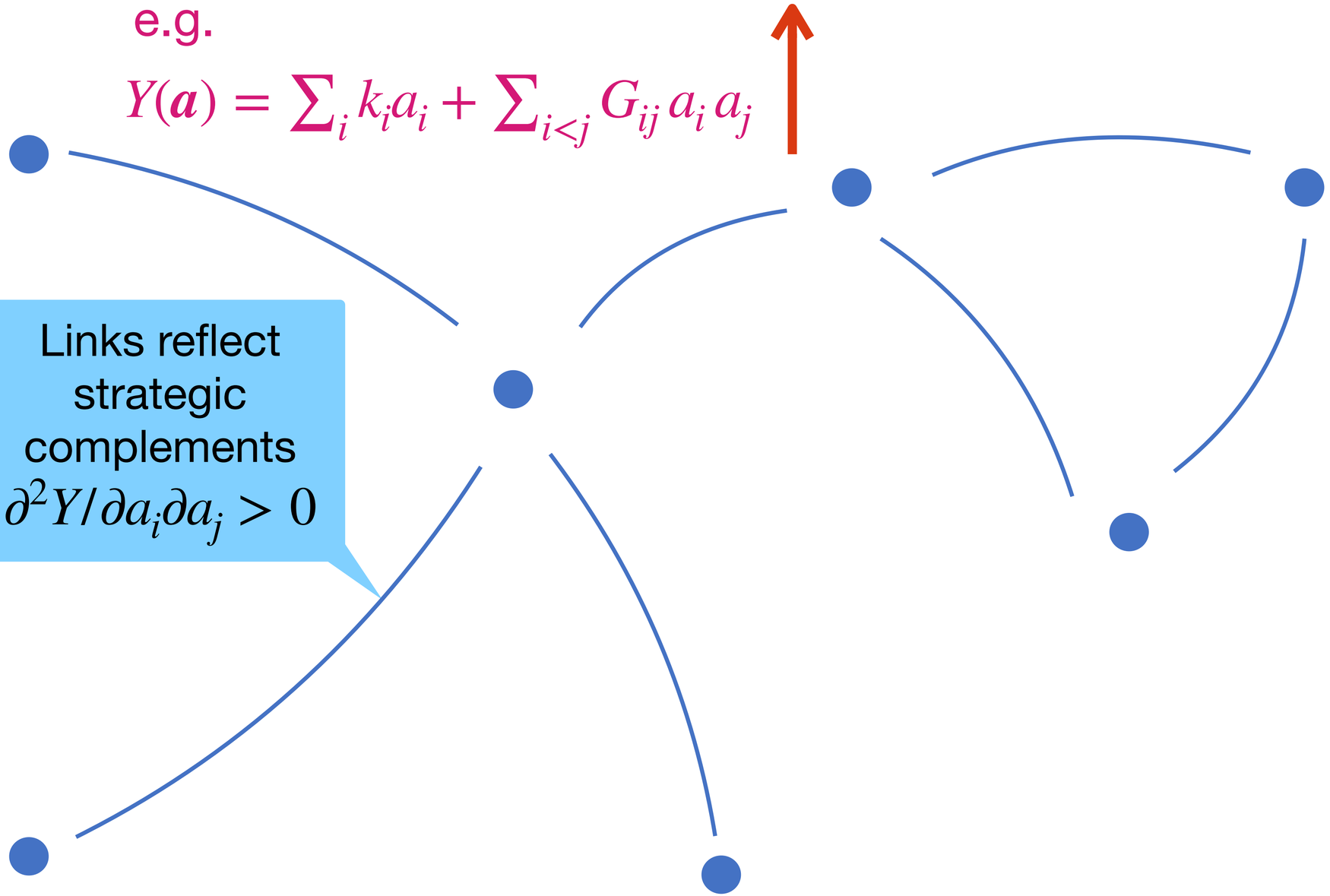
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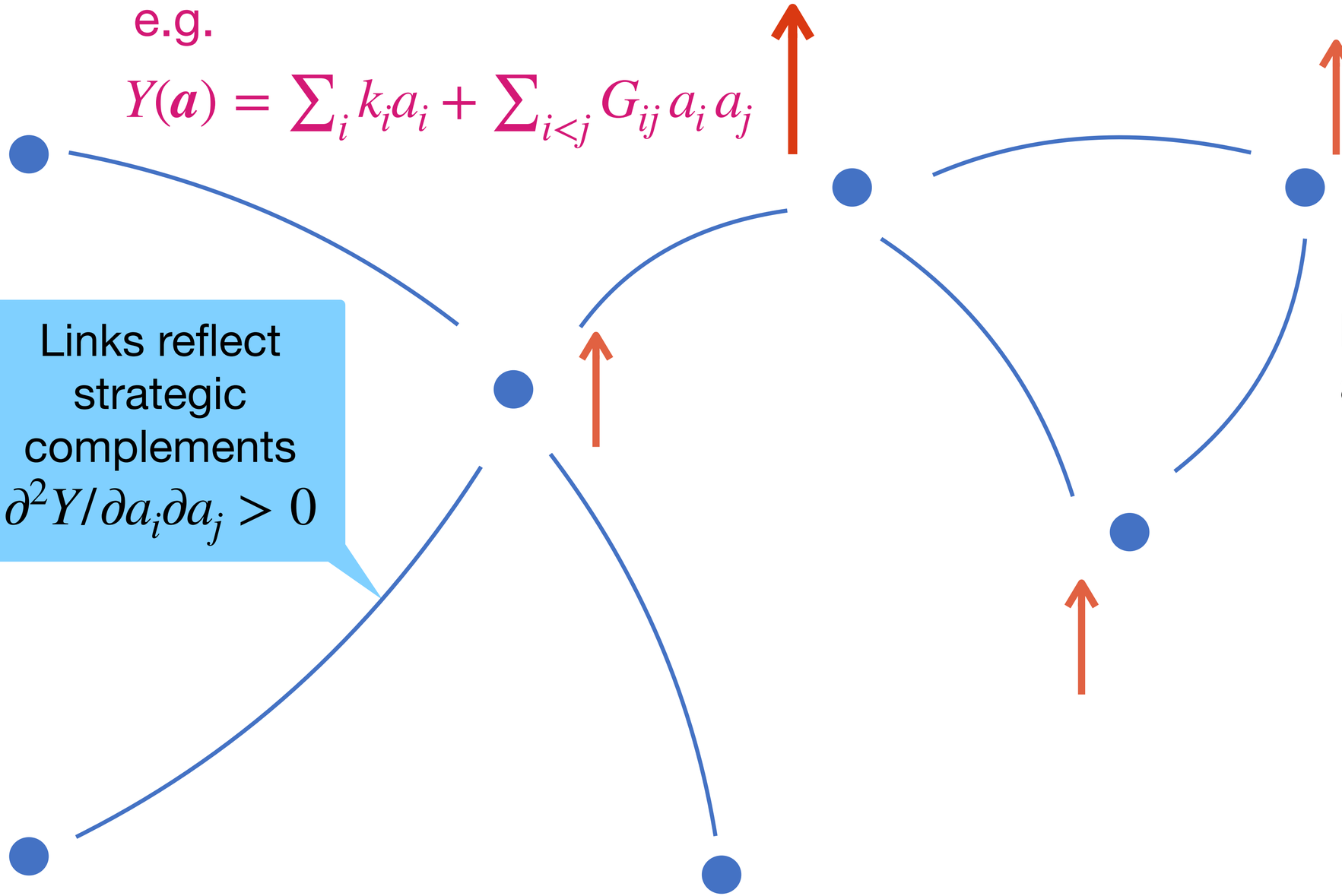
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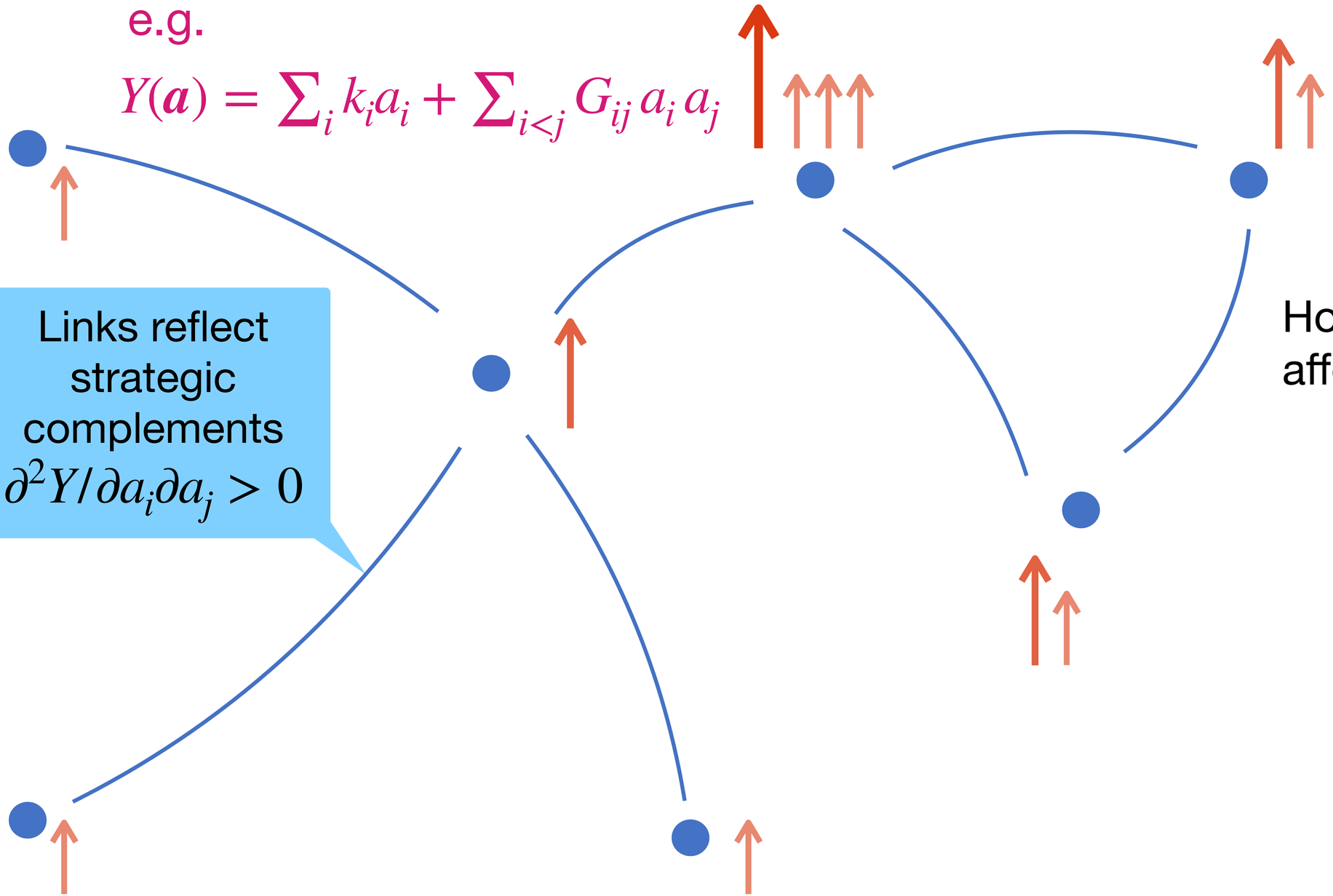
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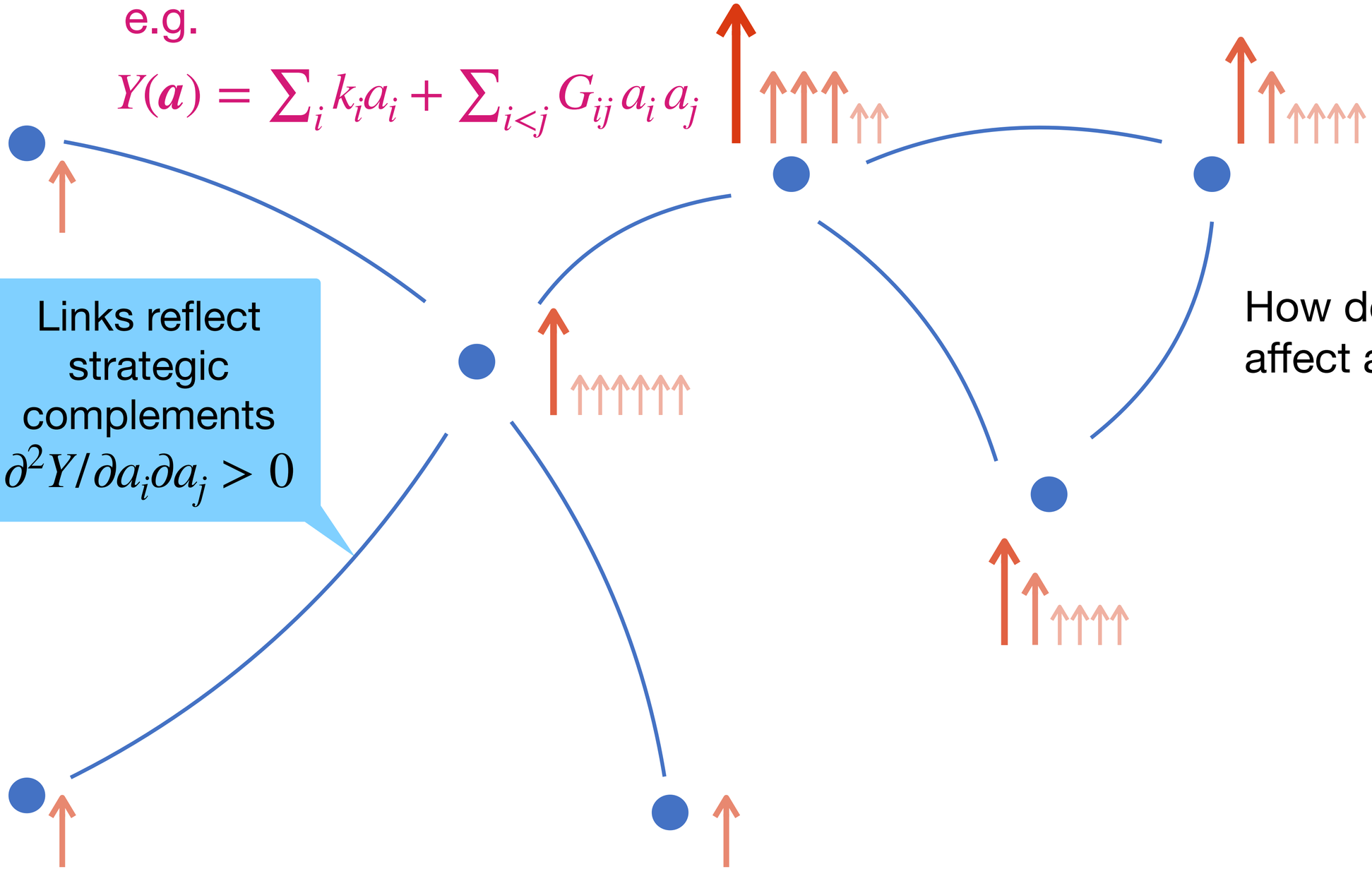
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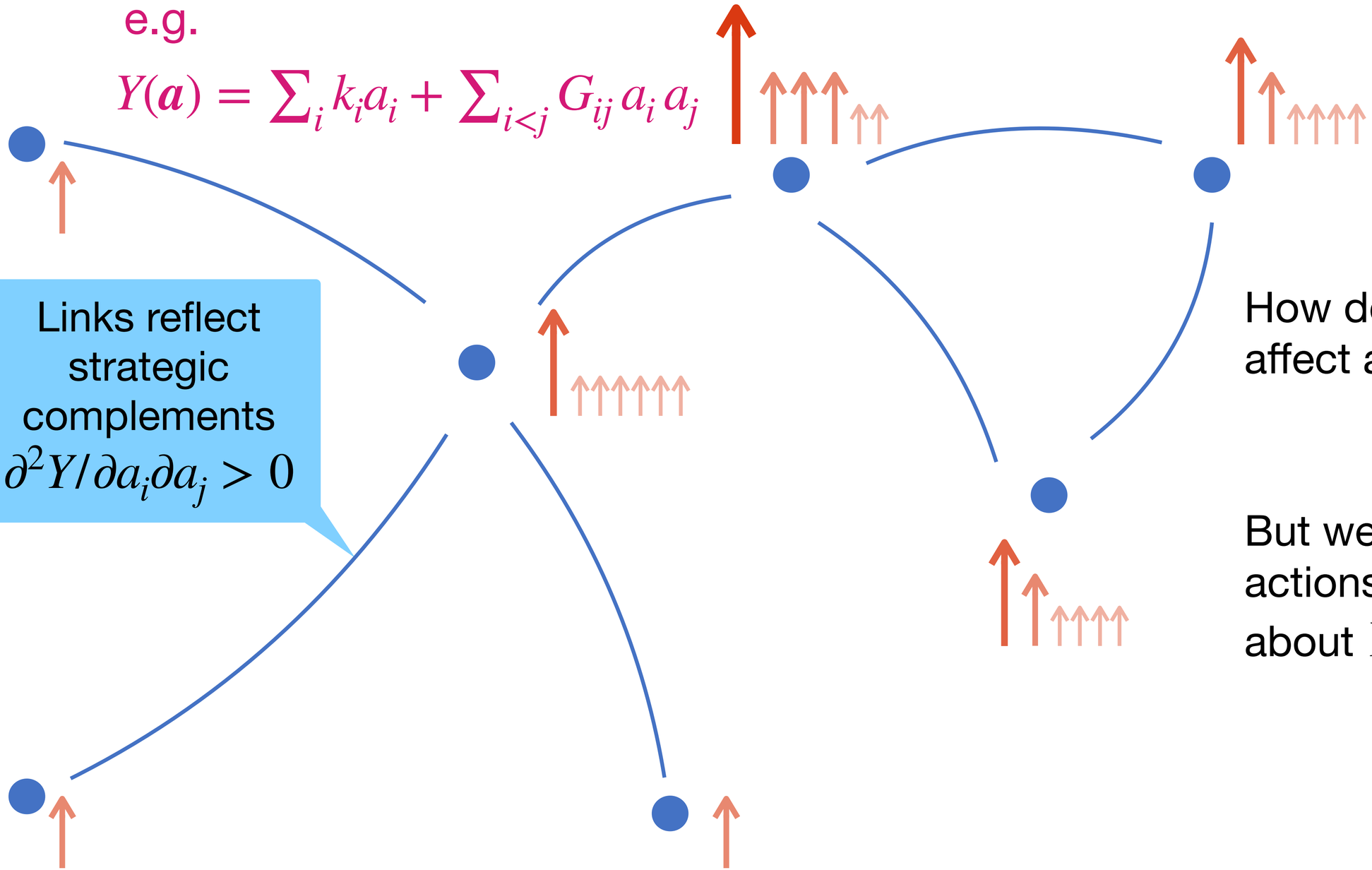
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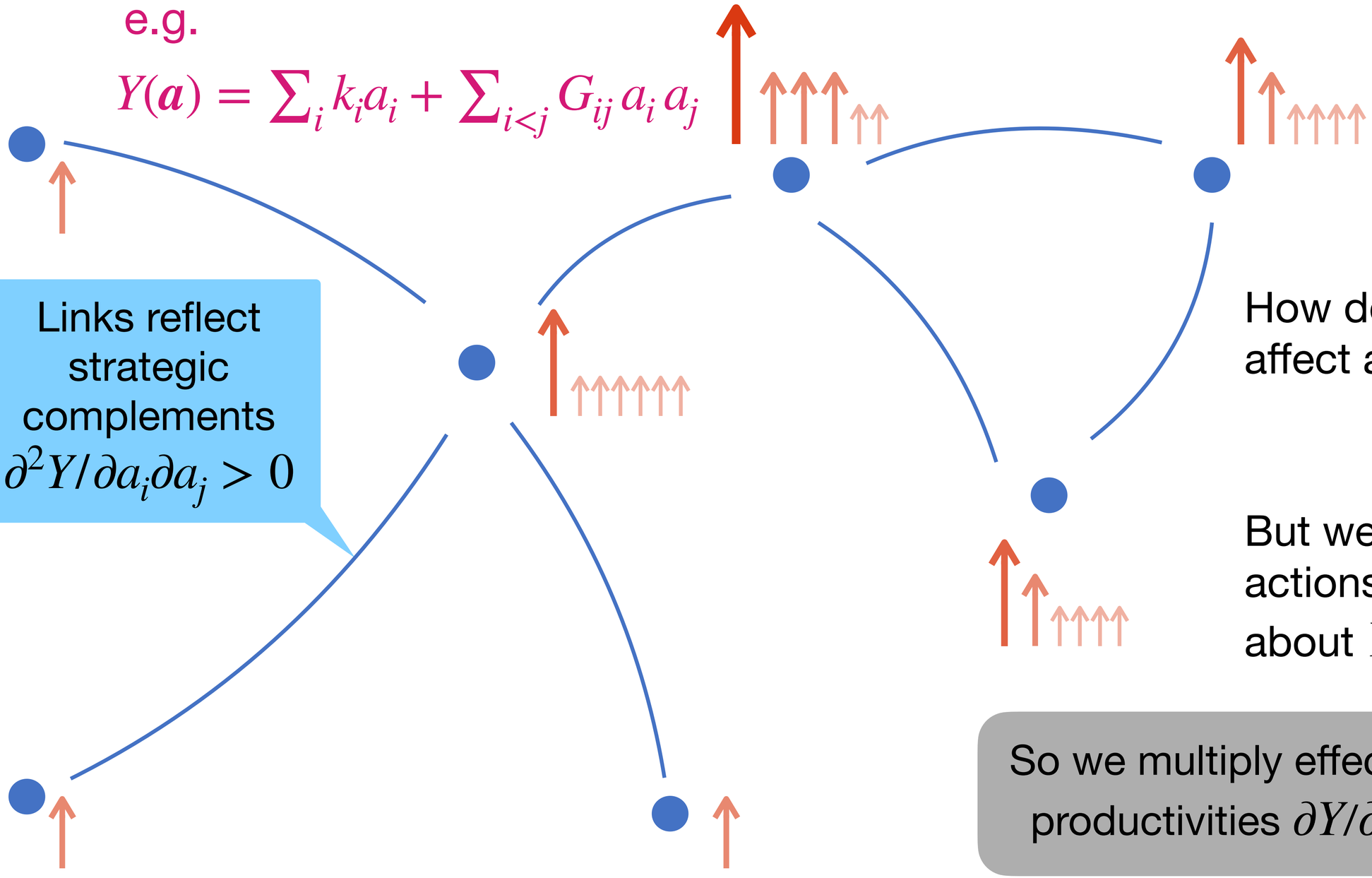
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But we don't care about actions per se, we care about $Y(\boldsymbol{a})$.

How an impulse to a player's δ_i propagates

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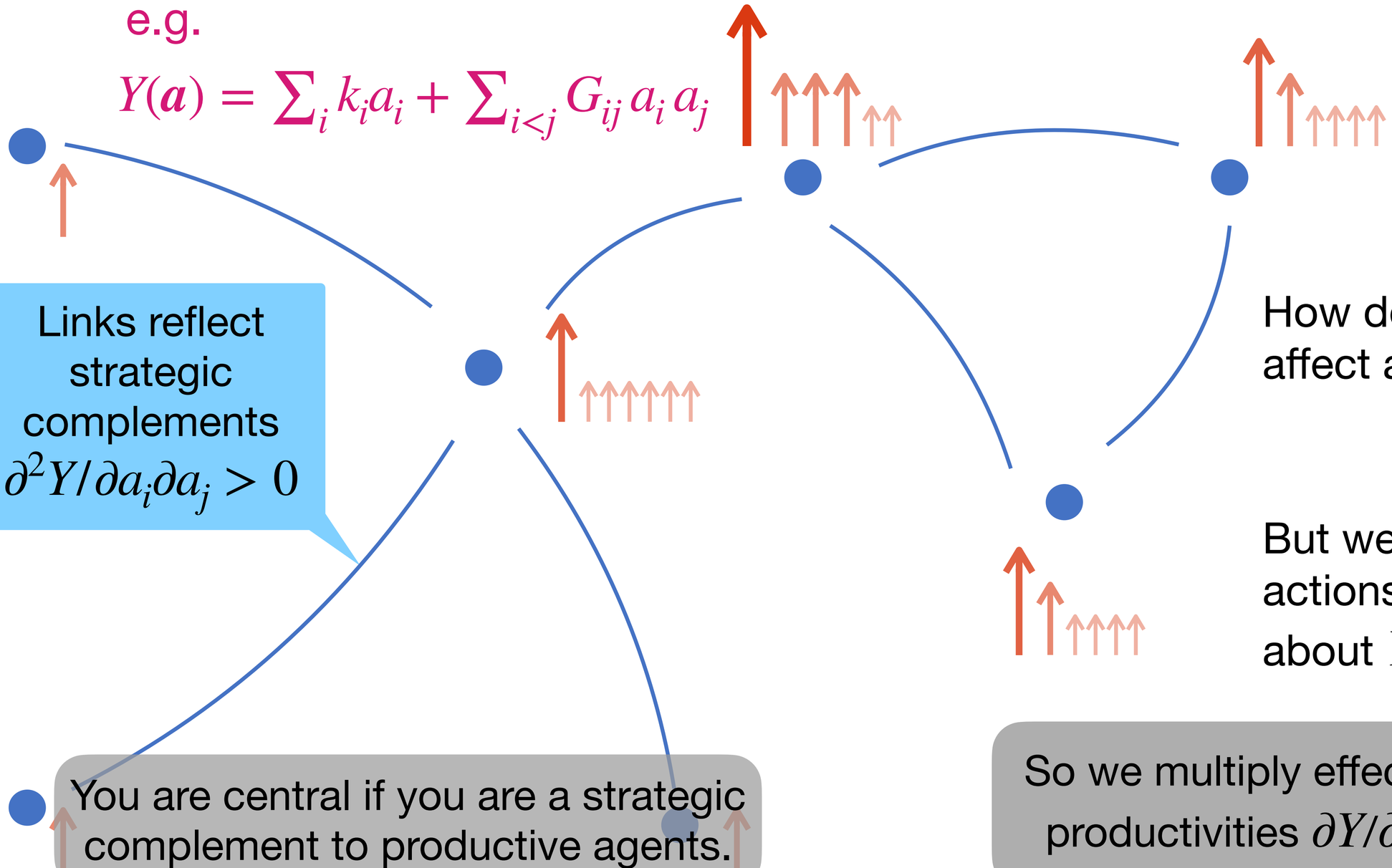
But we don't care about actions per se, we care about $Y(\boldsymbol{a})$.

So we multiply effects on j's actions by productivities $\partial Y / \partial a_j$ to get centrality

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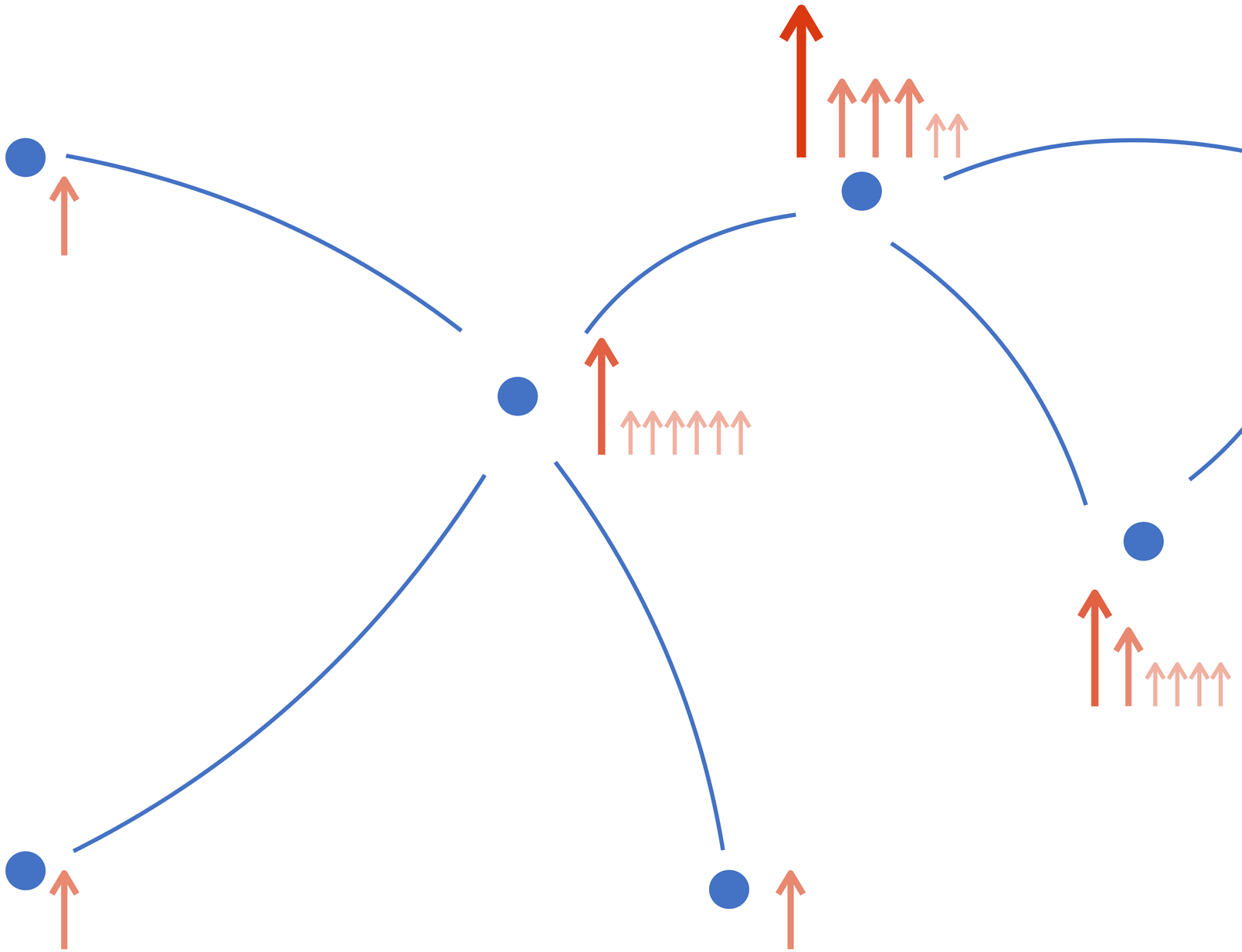
How does the impulse affect actions?

But we don't care about actions per se, we care about $Y(\boldsymbol{a})$.

You are central if you are a strategic complement to productive agents.

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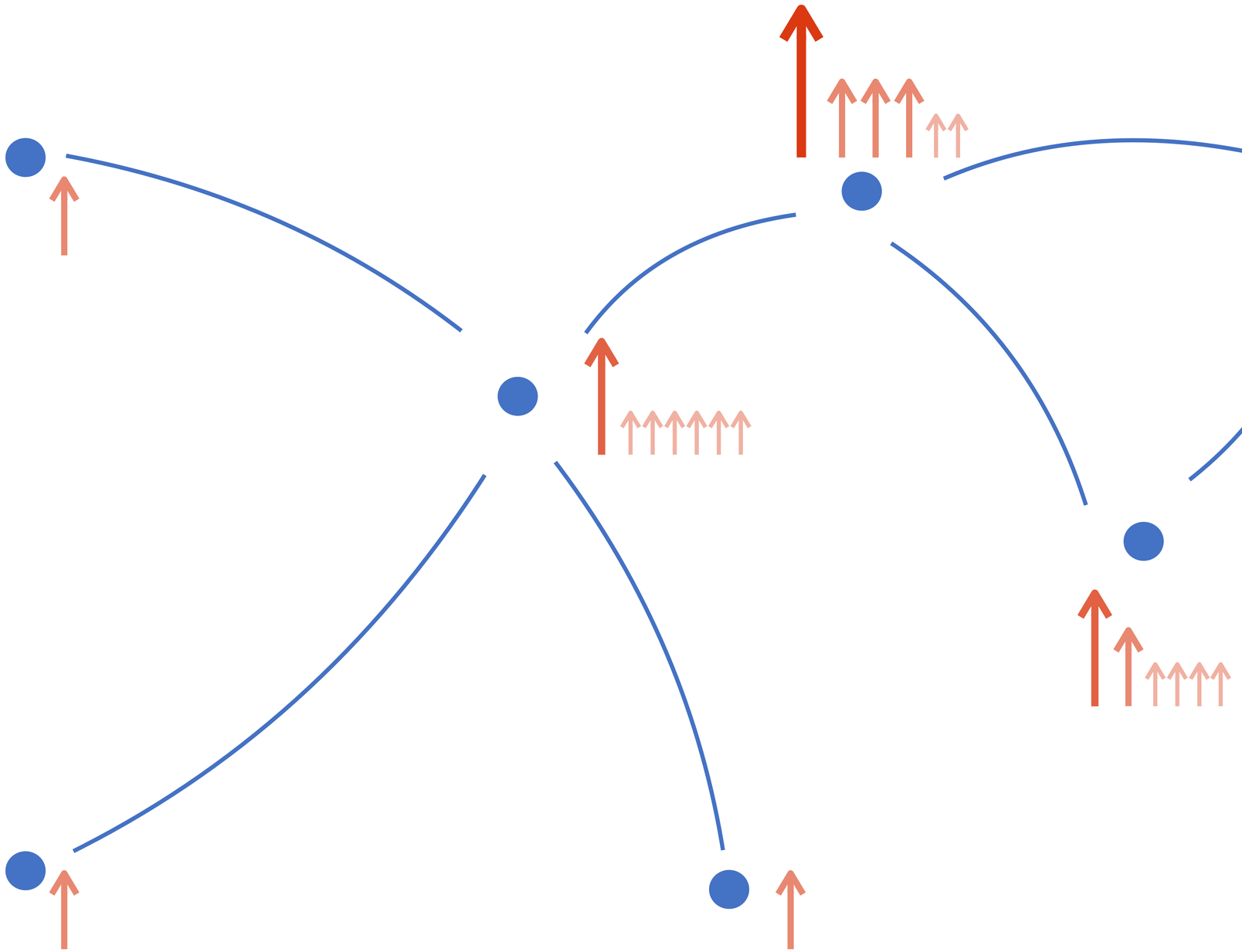
Turning this intuition into symbols

Define $G_{ij} = {}^{\partial^2 Y}/_{\partial a_i \partial a_j}$

$$W_{ji} = \tau_j G_{ji}$$

$$\boldsymbol{c}^\top := \boldsymbol{\alpha}^\top \left[I - P'(Y)\boldsymbol{W} \right]^{-1}$$

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Balance result in the binary case

Agent i's ***productivity*** is

$$\alpha_i = \frac{\partial Y}{\partial a_i}(\boldsymbol{a}^*)$$

Agent i's ***centrality*** is

$$c_i = \frac{\mathrm{d}Y(\boldsymbol{a}^*)}{\mathrm{d}\delta_i}$$

effect on Y of increasing i's marginal returns to effort

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Proposition. At an optimal contract $\boldsymbol{\tau}$ and associated principal-optimal eq'm $\boldsymbol{a}^*(\boldsymbol{\tau})$

$$\mu_i \cdot \alpha_i \cdot c_i$$

is equal across all agents with $\tau_i > 0$.

Balance result in the binary case

Agent i's ***marginal utility of money is***

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income-based responsiveness

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Corollary. If everyone has the same strictly concave utility for money, agents' positive payments τ_i are ranked according to $\alpha_i \cdot c_i$.

Balance result in general

Agent i's ***marginal utility of money is***

$$\mu_{i,s} = U_i'(\tau_{i,s})$$

income-based responsiveness

Agent i's ***productivity*** is

$$\alpha_i = \frac{\partial Y}{\partial a_i}(\boldsymbol{a}^*)$$

productivity-based responsiveness

Agent i's ***centrality*** is

$$c_i = \frac{\mathrm{d}Y(\boldsymbol{a}^*)}{\mathrm{d}\delta_i}$$

effect on Y of increasing i's marginal returns to effort

Proposition. At an optimal contract, at all outcomes s such that $P_s'(Y) > 0$,

$$\mu_{i,s} \cdot \alpha_i \cdot c_i$$

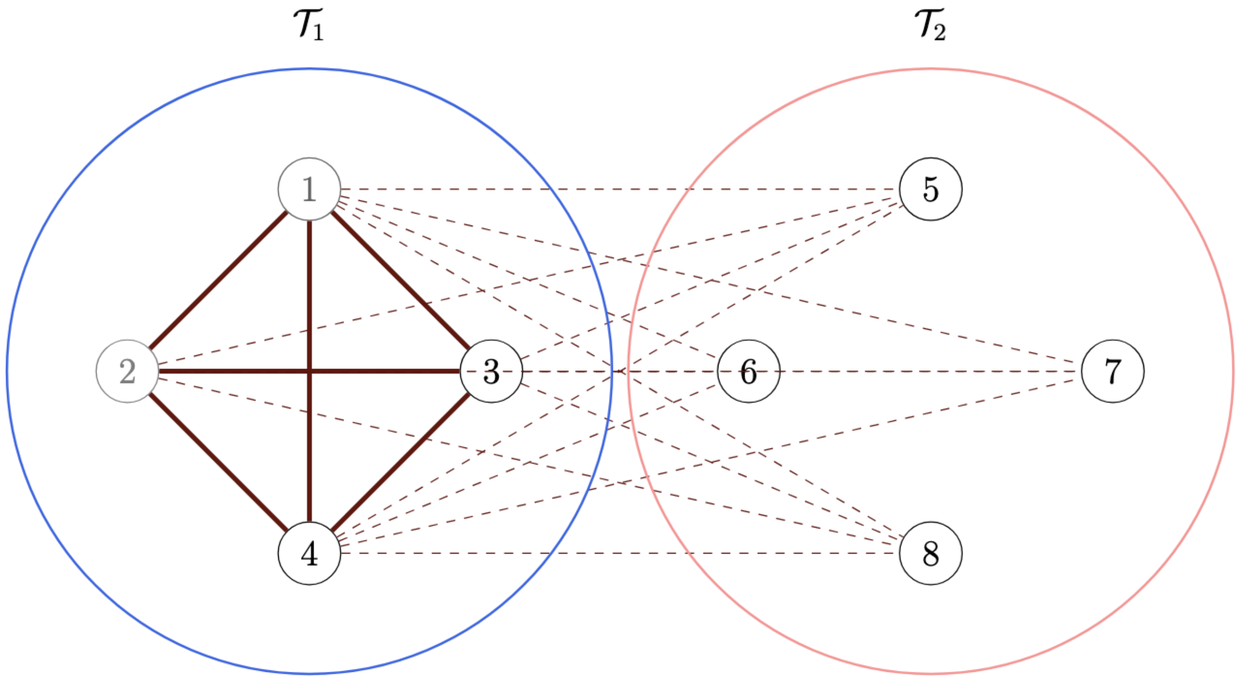
is equal across all agents with $\tau_{i,s} > 0$.

Lemma. At any eq'm, the product above is prop. to the principal's marginal returns of increasing i's payment in outcome s.

If this product were not equalized across agents, the principal could profitably reallocate bonuses

i.e. "hill-climb" in contract space.

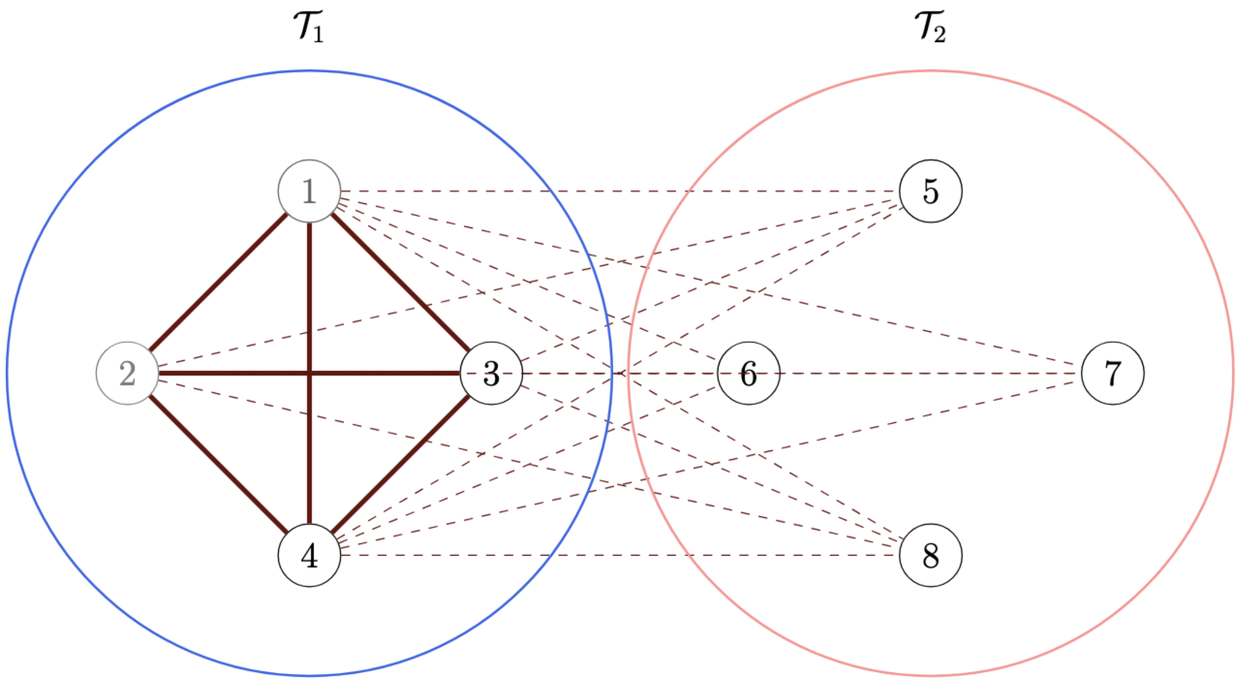
An intuition for why we care about centrality



very complementary
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very individually
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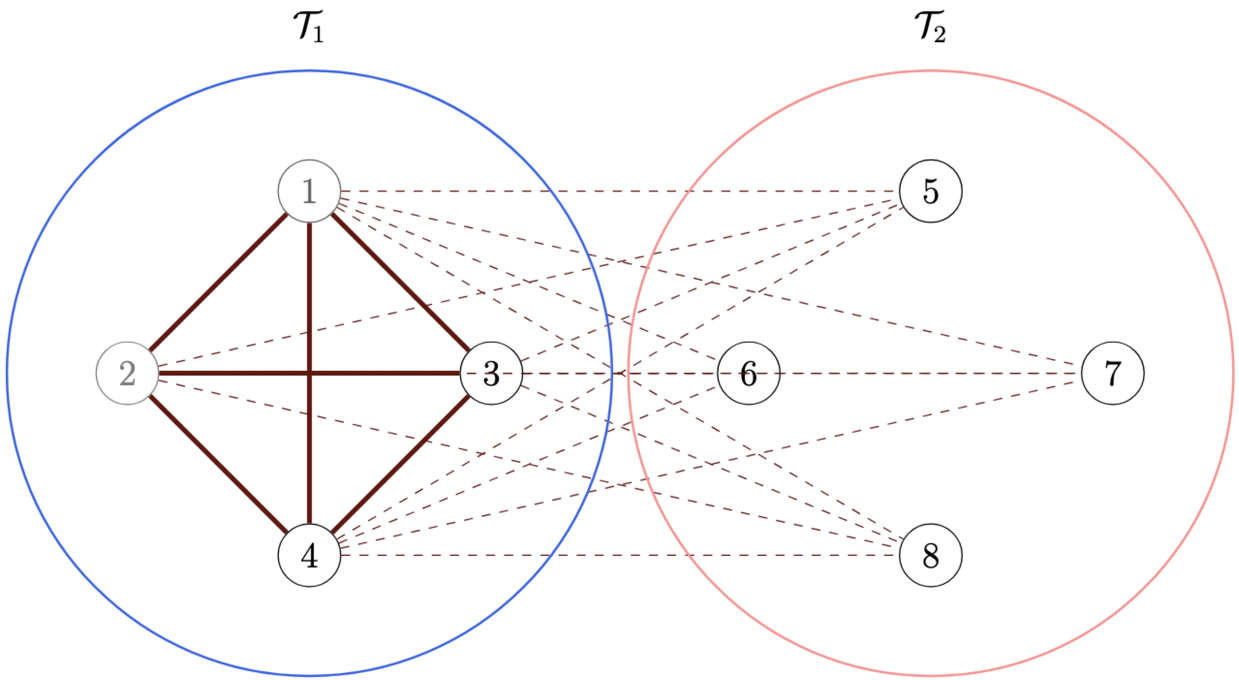
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$$Y(\boldsymbol{a}) = \sum_i k_i a_i + \sum_{i<j} G_{ij} a_i a_j$$

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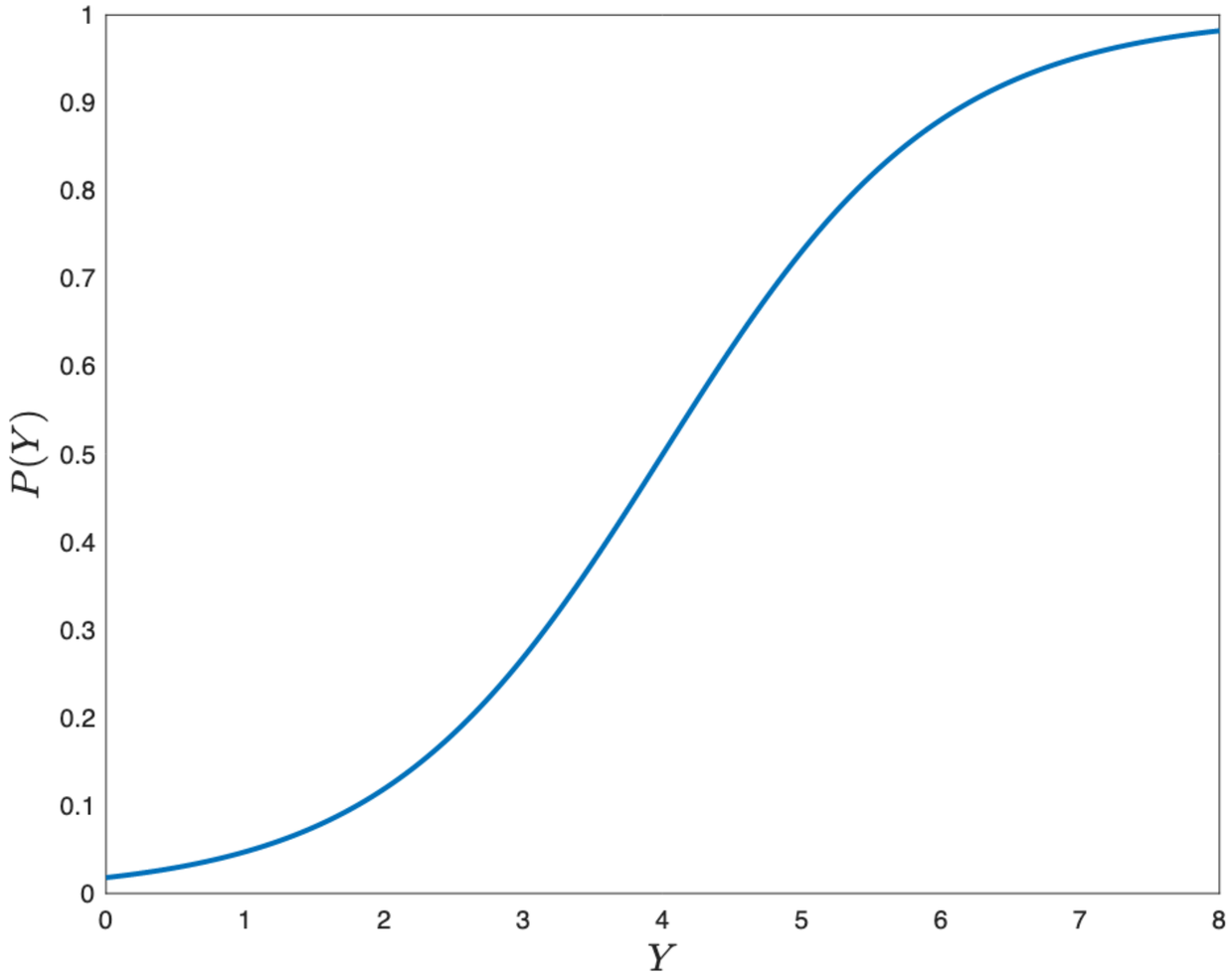
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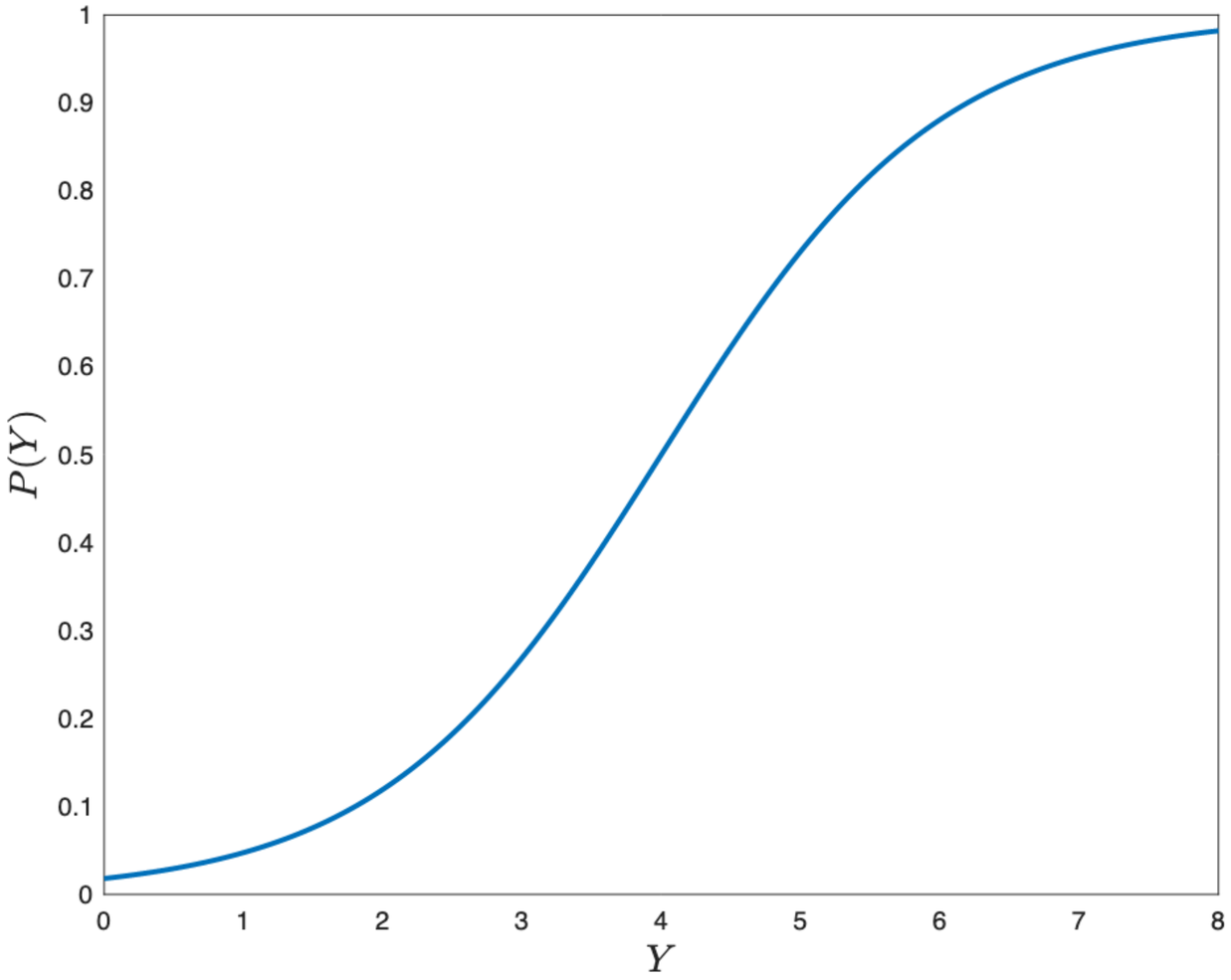
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We work out an example, show paying in proportion to standalone productivities gets 10% of optimal profit.

Application: More complementarity, less inequality

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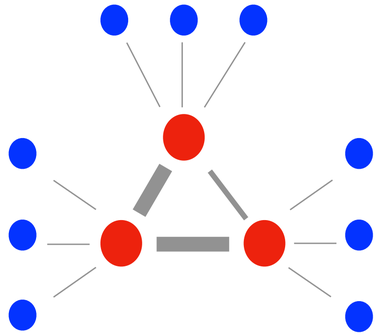
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Binary case with

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Example:
Core-periphery Network

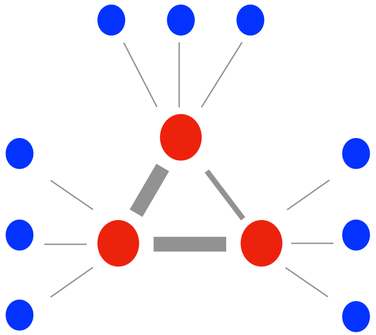


Binary case with

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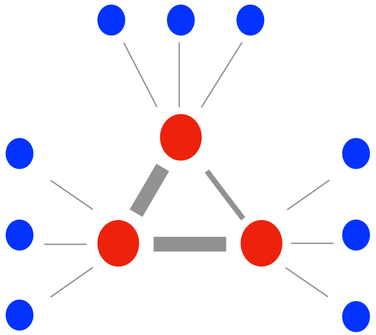
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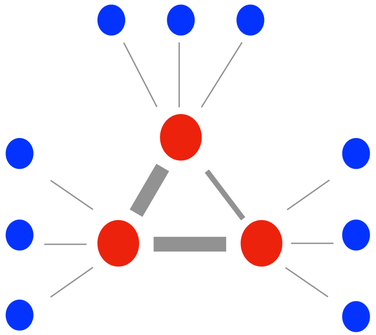
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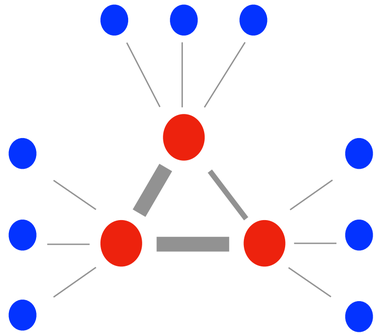
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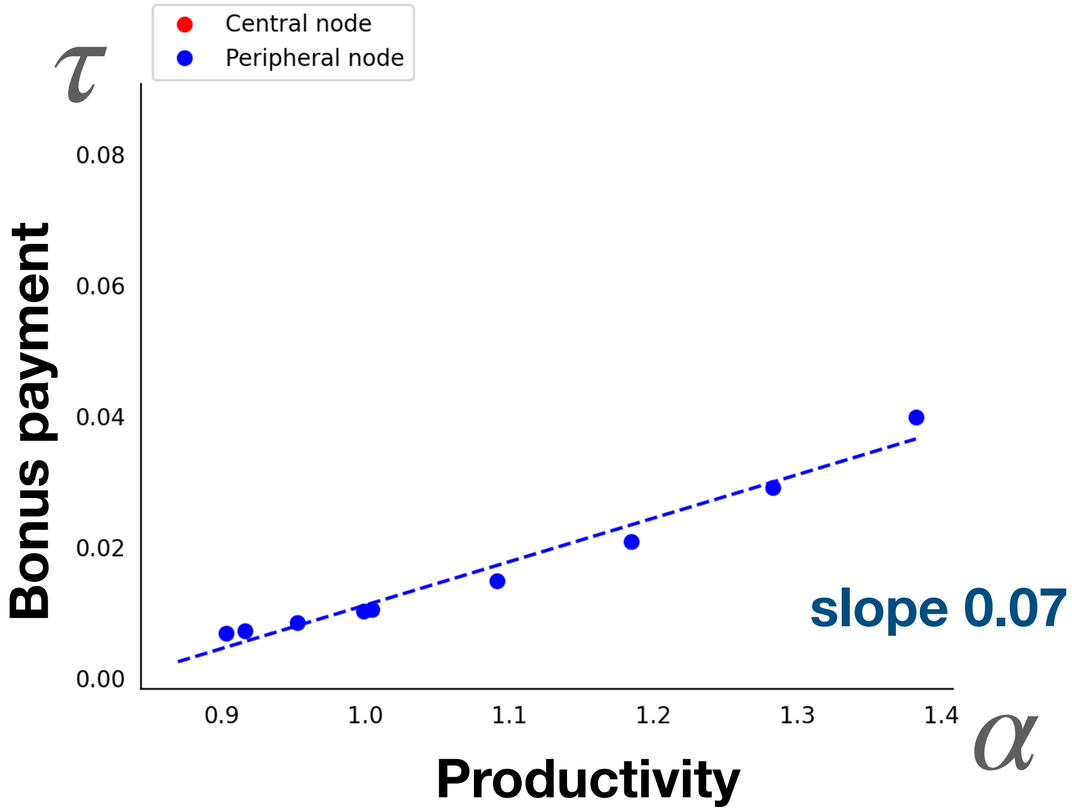
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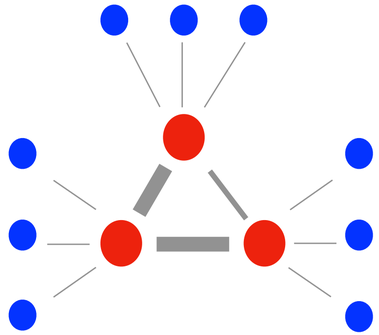
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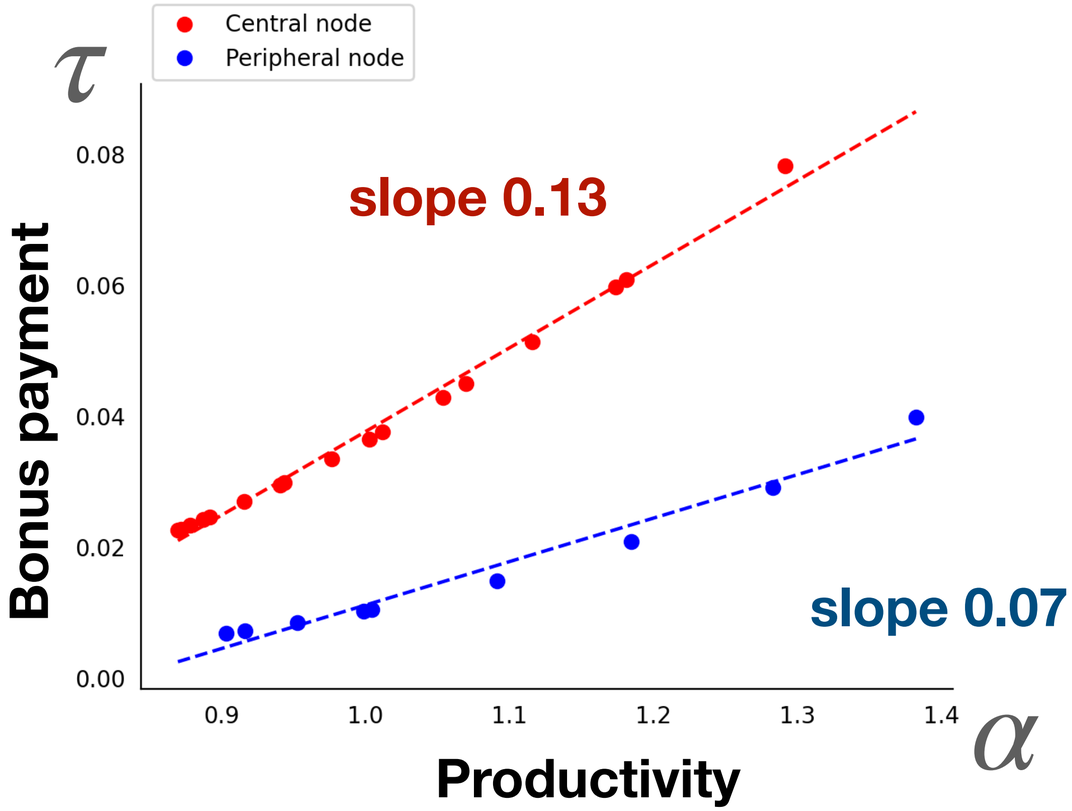
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Application: Indispensable tasks

Each agent performs one task that must succeed.

Only overall success is contractible. β_i measures task ability.

Task i succeeds with probability $q_i = 1 - e^{-\beta_i a_i}$.

Agent i earns share τ_i of the unit prize on success and pays cost $a_i^2/2$.

Tasks are independent, so project success is

$$Q(\mathbf{a}) = \prod_i q_i.$$

A failed task defeats the project. Greater reliability in one task raises the return to effort on every other task.

More capable agents receive less equity

At a profitable optimum, ability ranks agents' shares.

$$\beta_i > \beta_j \quad \Longrightarrow \quad \tau_i^* < \tau_j^*$$

The more capable agent also exerts less effort: $a_i^* < a_j^*$.

Yet the more capable agent's task succeeds more often: $q_i^* > q_j^*$.

A higher β makes each unit of effort more effective. The principal can assign that agent a smaller share while obtaining a more reliable task.

Higher ability lowers every equity share

Agent	Before ability	After agent 1 improves ability	Before equity share	After agent 1 improves equity share
1	5	8	25%	18%
2	5	5	25%	23%
3	5	5	25%	23%

With three indispensable tasks, every agent's success share falls.

The other agents work harder because the first task is more reliable.

Overall project success and the principal's profit rise.

Adding a required task raises equity costs

Equal ability $\beta = 10$; fixed prize. Each added task lowers success and principal profit.

Tasks	Equity per worker	Principal equity	Success
3	**13%**	61%	52%
4	**14.5%**	42%	37%
5	**17%**	14%	21%

Each existing worker gets a larger share and supplies less effort.

Equity and the cost of reliability

$B_i(q_i)$: expected payment needed to induce task reliability q_i.

$$\tau_i = \frac{B_i(q_i)}{Q} = \frac{B_i(q_i)/q_i}{B_i'(q_i)} = \frac{\text{average incentive cost}}{\text{marginal incentive cost}}$$

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Three implications of buying reliability

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- A teammate's failure acts like a tax on your incentive. Better teammates make your bonus more likely to pay, so you work harder despite a smaller share.
- An extra required task adds another way for everyone's bonus to disappear. Existing workers' effort becomes less valuable and their bonuses less effective. They receive larger shares but work less.

Why higher ability receives less equity

Risk-neutral agents with quadratic effort costs.

Performance is project success.

The principal's contract FOC:

$$0 = -Q + (1 - \sum_j \tau_j) \underbrace{\alpha_i}_{\text{productivity}} \underbrace{c_i}_{\text{centrality}}$$

Centrality captures effort responses to payoff shocks:

$$\alpha_i = \frac{\partial Q}{\partial a_i} = \frac{\beta_i Q}{e^{t_i} - 1}$$

$$c_i = \frac{dQ}{d\delta_i}, \quad \text{shock} : + \delta_i a_i$$

$$\tau_i Q = \underbrace{\frac{t_i(e^{t_i} - 1)}{\beta_i^2}}_{\text{worker FOC}}$$

Combining the contract and worker FOCs:

$$t_i = \beta_i a_i$$

$$\beta_i^2 Q = \underbrace{(e^{t_i} - 1)[(1 + t_i)e^{t_i} - 1]}_{D(t_i)\uparrow}.$$

The resulting optimal share:

$$\tau_i = \underbrace{\frac{t_i}{(1 + t_i)e^{t_i} - 1}}_{w(t_i)\downarrow}$$

At common project success, higher ability raises effective effort. The declining share rule gives that agent less equity.

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These conditions jointly ensure each agent's utility function is strictly concave in own action, guaranteeing that FOC characterize a unique equilibrium

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- should be relaxed — interesting technical questions!

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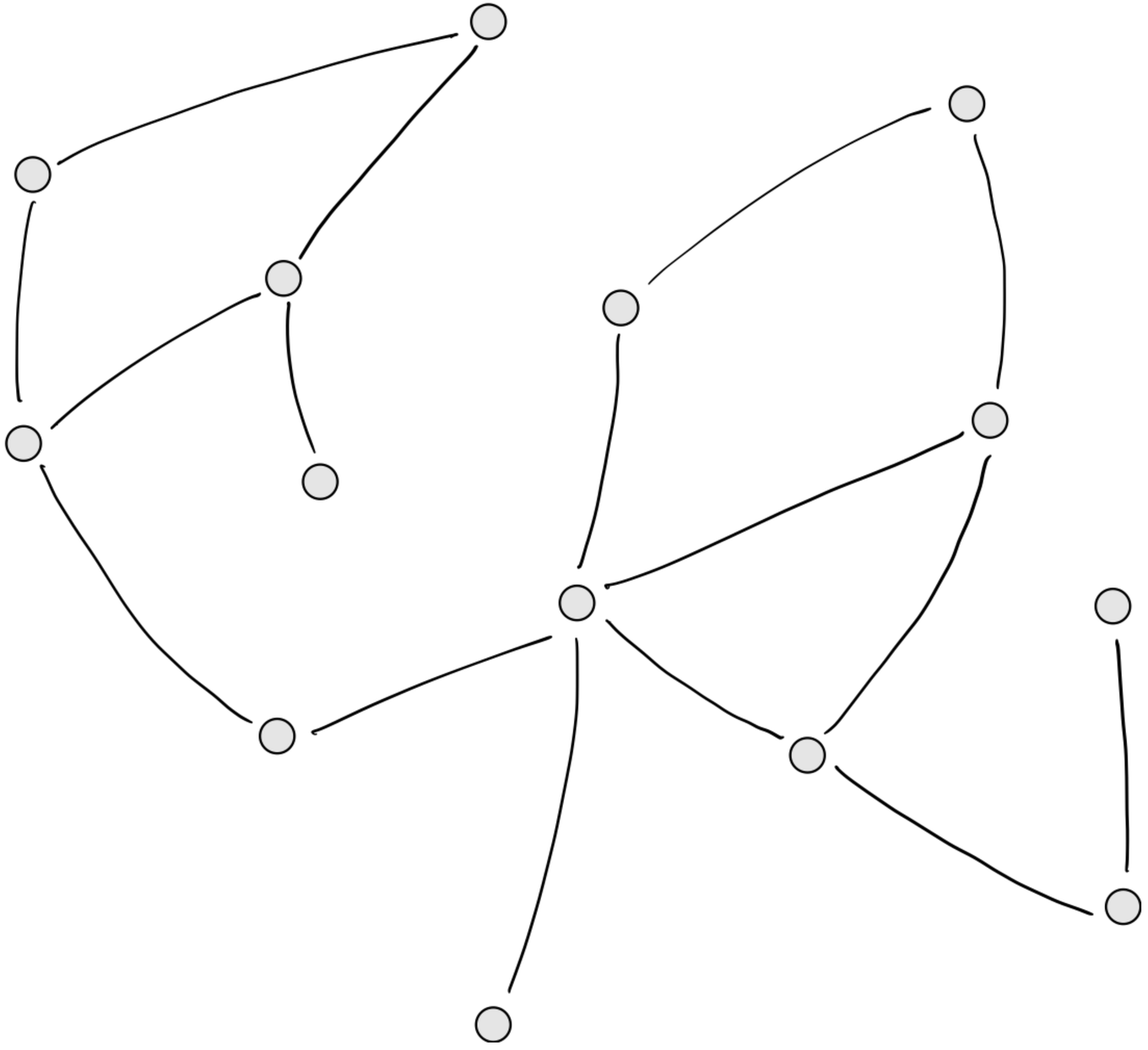
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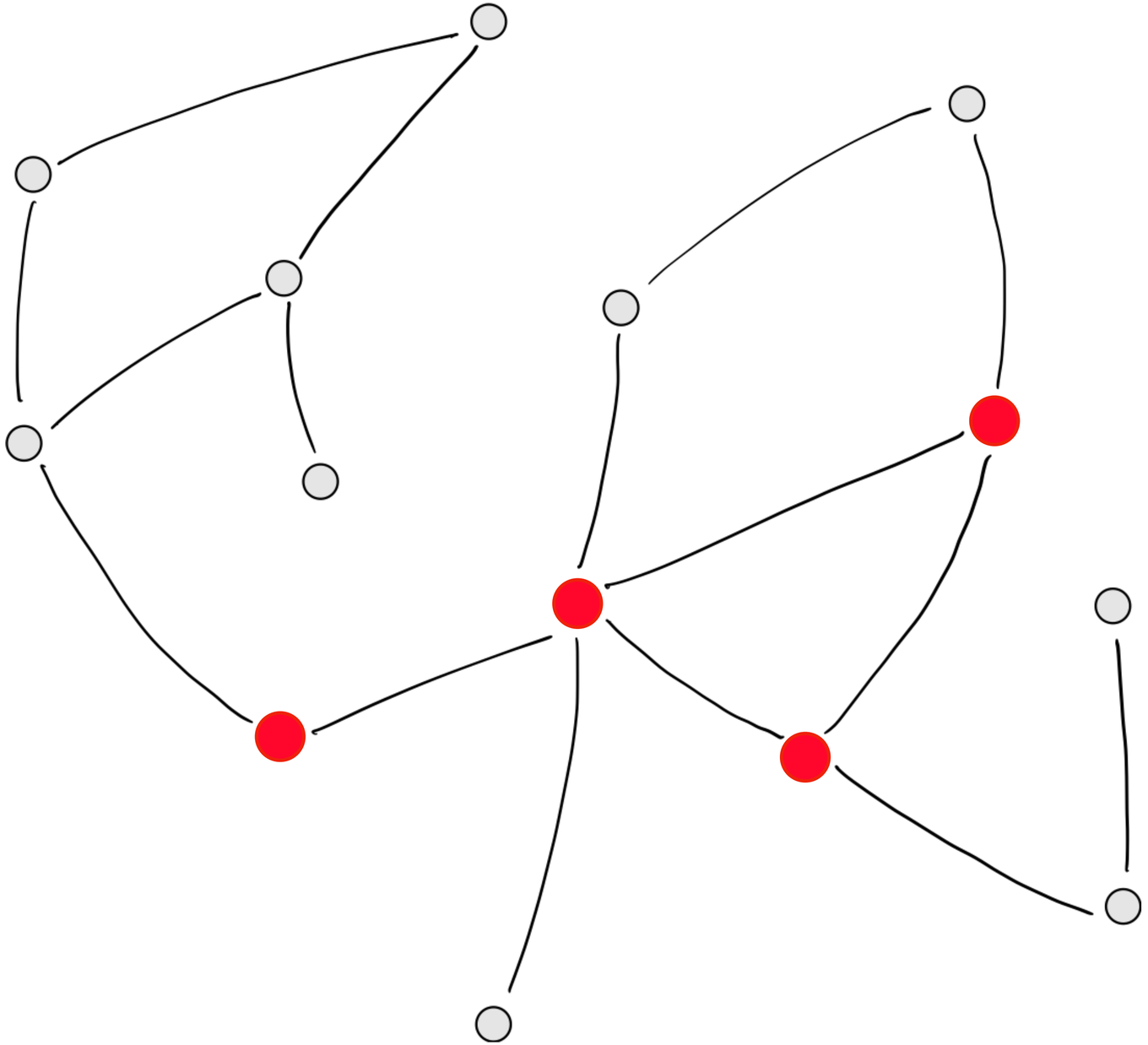
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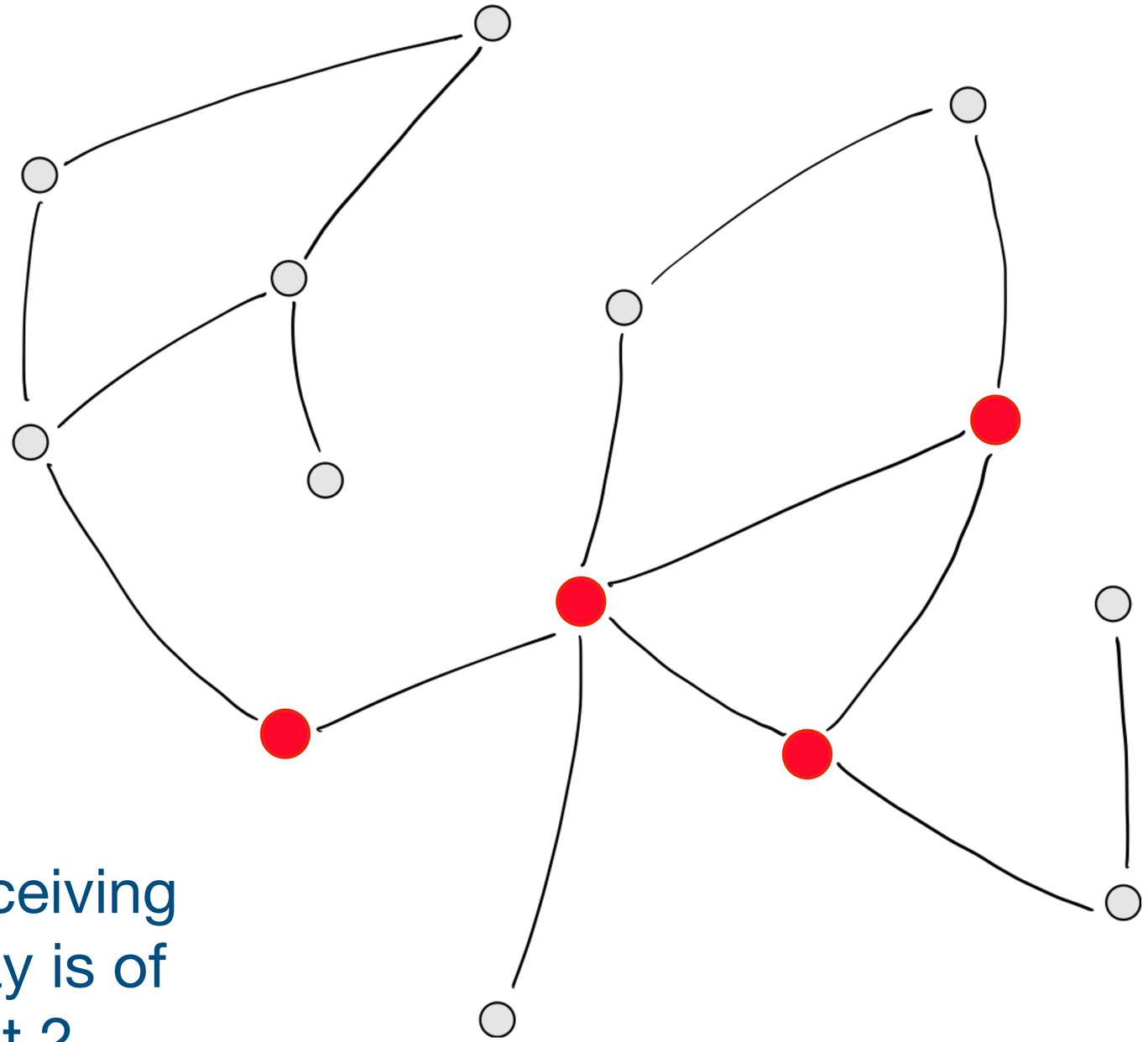
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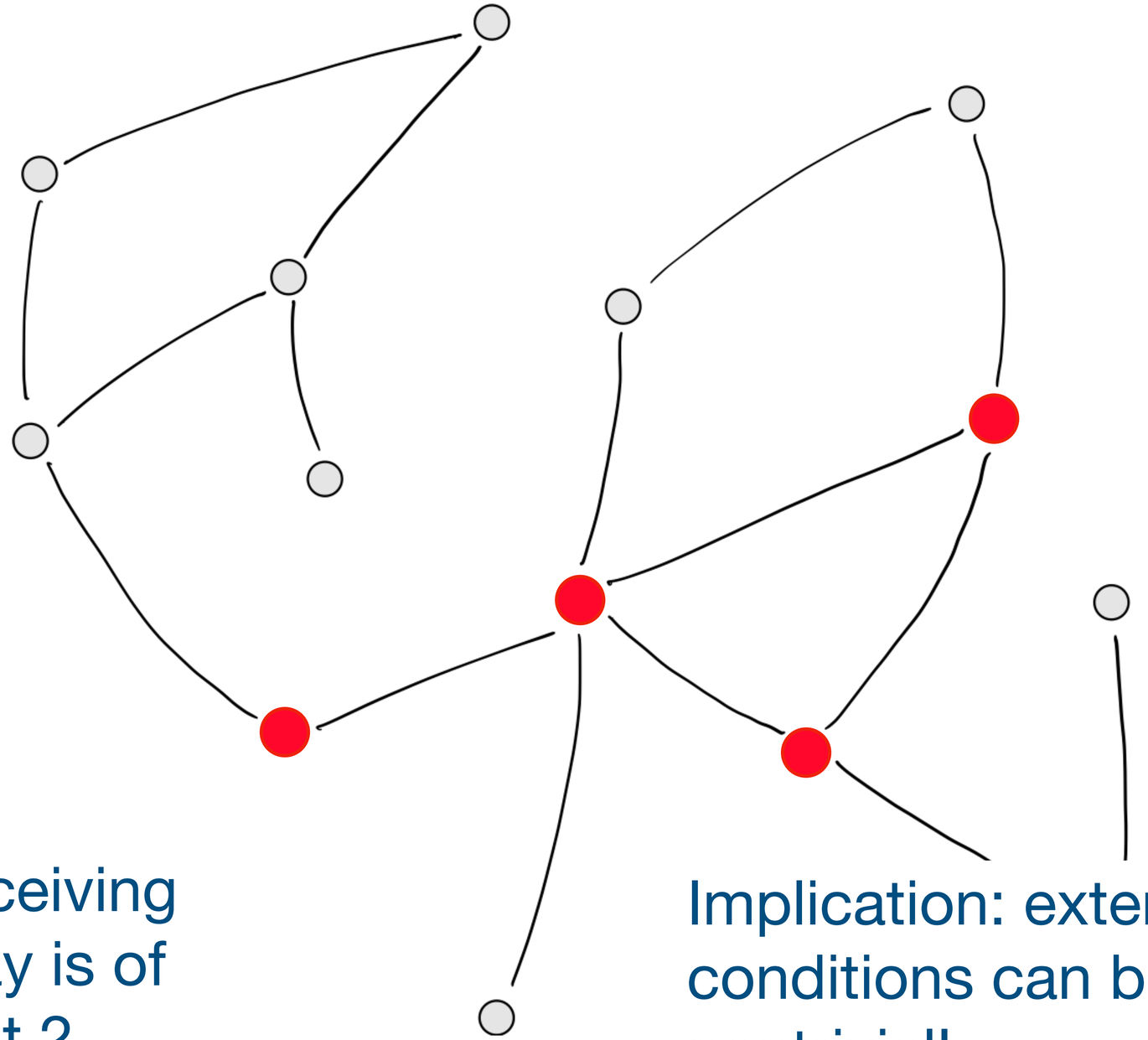
The network on them induced by G has diameter at most 2.







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Implication: extensive-margin conditions can be highly nontrivial!

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Statistical problem

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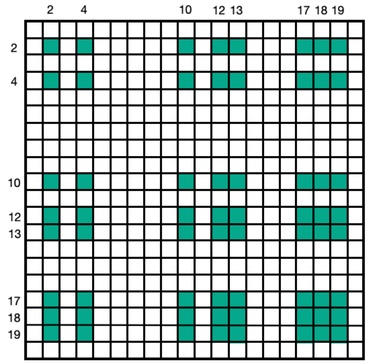
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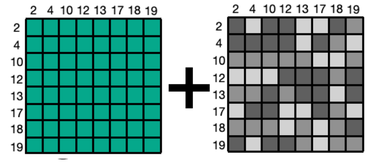
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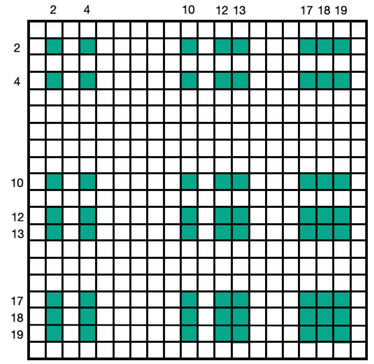
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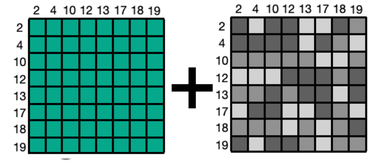
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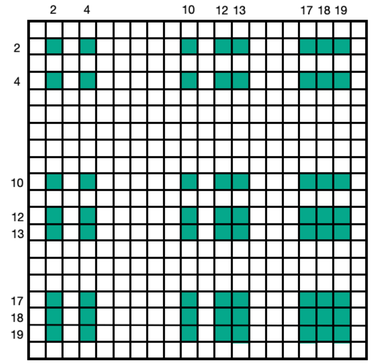
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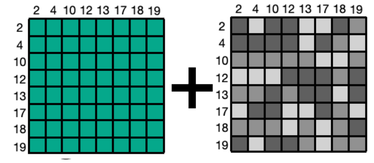
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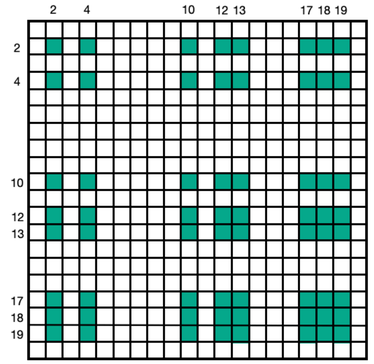
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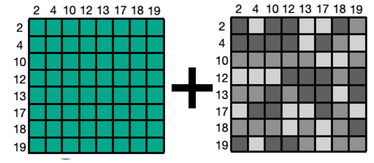
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Take advantage of large matrix statistics — robust recovery of certain eigenvectors (cf. matrix completion).

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To assess local optimality or locally improve, it suffices to **approximate game by quadratic game**, use network game theory.

Summing up

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MU of $ $\times$ **productivity** $\times$ **centrality**

Other things we do

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An agent who becomes more productive can be forced to become less central.

Counterintuitive: Increasing productivity can *decrease* centrality

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Two agents, risk-neutral:



$$\mathcal{U}_i(a_i, \tau) = \tau_i - a_i^2/2.$$

Quadratic team performance:

$$Y(\boldsymbol{a}) = \sum_i k_i a_i + \beta a_1 a_2$$

Binary setting, $P(Y)$ given.

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At $k_1 = k_2 = 1$ principal gives equal incentive contracts to both

If k_1 is perturbed slightly upward, then at the optimal contract $\alpha_1 > \alpha_2$ and $c_1 < c_2$.

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Moral hazard, e.g.:

Holmstrom (79), *Moral hazard and observability*

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Our contribution: endogenize incentives with nonparametric team production function.

We leverage ideas from this literature without the strong parametric assumptions typical in network games.

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A test of whether incentives are optimal.

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Corollary (general model). If everyone has the same strictly concave utility for money, agents' positive payments $\tau_{i,s} > 0$ are ranked the same **in every outcome**.

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First and second derivatives of the primitives (e.g. Y and the U_i) are sufficient to calculate these.